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Do coupled population, economic, and land use dynamics explain household energy transitions in Malawi?

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ABSTRACT

In sub-Saharan Africa, population growth coupled with weak or missing markets for clean energy means that the absolute number of people relying on woodfuels for cooking is growing. Charcoal, which has distinctly different environmental and economic implications from firewood, has become one of the most consumed fuels in the region. However, among rural populations, the drivers and consequences of charcoal use remain under-studied. We examine rural woodfuel dynamics in Malawi, a majority rural country with high poverty, rapid population growth, lagging economic development, and abundant but rapidly declining forest resources. We examine how household energy transitions are influenced by concurrent demographic, economic, and land use changes. Using four waves of national household surveys and satellite-derived forest cover datasets, we investigate trends in household energy consumption, identify factors driving rural household fuel choices, and characterize geographic patterns of forest cover loss during 2004–2020. During this time, the share of rural households using charcoal grew from 3.9% to 13.4%. Rural households that are smaller, headed by younger individuals, and are asset-rich but land-poor had greater odds of using charcoal compared to firewood, indicating that rural livelihood dynamics have important consequences for household energy transitions. Several protected areas have experienced > 20% forest cover loss since 2001, characterized by geographic patterns consistent with unsanctioned woodfuel extraction. These findings emphasize a renewed need for integrated natural resources, energy, and livelihoods management strategies. Our insights provide information for policy makers to better predict and understand the continued growth of charcoal's popularity in Malawi and similar contexts.

1. Introduction

Fifty years ago, the development community rallied around speculation of a household energy crisis in low-income countries where populations relied heavily on woodfuels (mainly firewood and charcoal) for cooking and heating energy. The overarching hypothesis was that rapid population growth was a primary driver of unsustainable woodfuel consumption, potentially leading to rapid and irreversible deforestation and leaving large populations without a source of household cooking and heating energy (Arnold et al., 2006; Eckholm, 1975). The "woodfuel crisis" hypothesis was subsequently laid to rest through two major contributions to our understanding of household energy dynamics in low-income countries. First, on the supply side, we have a better understanding of the contribution of firewood and charcoal production in processes of deforestation and forest degradation (Bailis et al., 2015; Dewees, 1989; Jagger & Kittner, 2017; Masera et al., 2015). Even at relatively high population densities, apart from a few unique settings, reliance on woodfuels for household cooking has been a minor contributor to deforestation and forest degradation historically (Allen & Barnes, 1985; Bailis et al., 2015). Second, on the demand side, many low-income countries have avoided the hypothesized woodfuel crisis by progressing through concurrent economic growth, demographic

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changes including urbanization and reduced fertility rates, and transitions to modern cooking fuels and technologies (Cline-Cole et al., 1990; Liao et al., 2021). Since 1990, the share of households reliant on woodfuels has decreased significantly in several countries that have undergone concurrent demographic and economic transitions, including India (89% to 29%), China (67% to 17%), South Africa (57% to 12%), and Brazil from (24% to 4%) (Global Health Observatory, 2024).

However, many countries in sub-Saharan Africa (SSA) are on a divergent energy transition trajectory towards greater biomass reliance, reanimating concerns about the demand and supply dynamics of woodfuels (Stoner et al., 2021). Dramatic population growth combined with weak or missing markets for clean cooking fuels has led to a growth in the absolute number of people reliant on woodfuels in the region (Rose et al., 2022). Furthermore, a unique feature of energy trends in SSA, and one that partially explains its divergent energy consumption trajectory, is the emergence and persistence of charcoal as a widely popular cooking fuel. Approximately 65% of the global charcoal supply for household cooking is produced and consumed in SSA (Mensah et al., 2022). For decades, urban dwellers have reported a preference for charcoal over firewood for cooking, due to several advantageous characteristics: availability year-round; ease of transport and trade due to lower weight per energy unit compared to firewood; and limited smoke when burned which is appealing for use in dense urban environments (Idowu et al., 2023; Kambewa et al., 2007; Zulu, 2010; Zulu & Richardson, 2013). Still, charcoal emits high levels of carbon monoxide and particulate matter, failing to improve over firewood in terms of impacts on human health and environmental degradation (Idowu et al., 2023; Lee et al., 2020; McCord et al., 2024). Increased woodfuel reliance therefore remains an issue of great environmental and public health concern.

Several other factors contribute to the rapid rise in charcoal reliance in the past two decades. First, a lack of reliable electricity supply in the region has forced households to continue to rely on polluting fuels for cooking (Rose et al., 2022; Smith et al., 2017), highlighting the influence of macro-level economic processes on household energy consumption dynamics. Second, alternative fuels with fewer negative impacts on human health and environment, such as liquid petroleum gas, are not yet widely available throughout SSA due to weak or missing supply chains and constraints in financial accessibility to households (Pachauri et al., 2021; Richardson et al., 2025). Unreliable electricity supply and weak or missing supply chains for clean fuels are well established factors influencing energy transitions (or lack thereof) and most relevant to rapidly growing urban populations. In this paper, we identify a third reason that explains charcoal's rise to popularity in SSA by carefully examining rural woodfuel consumption dynamics. In the majority of SSA countries, prevalence of rural charcoal reliance rose rapidly from near 0% in the early 2000 s to a range of 5–20% as of the early 2020 s, shifting the share of households reliant on firewood to charcoal (Rose et al., 2022; Stoner et al., 2021). Given that most people in SSA reside in rural areas, the rural charcoal user base approaches the size of the urban charcoal consumer base in some countries. Notably, Stoner et al. (2021) estimated that approximately half of households reliant on charcoal in Kenya, Rwanda, Malawi, and Uganda are rural (Global Health Observatory, 2024). Despite this, the reasons for the increase in rural charcoal consumption is underexamined in both scholarship and policy implementation.

Malawi is an important case for understanding the woodfuel energy transition (from firewood to charcoal) in settings characterized by limited economic growth, rapid population growth, urbanization, and high rates of deforestation. Malawi's population has one of the highest poverty rates in the world at 51%, an 82% rural population, and is over 95% reliant on woodfuels across both urban and rural populations (National Statistical Office, 2021; United Nations, 2022). The Malawian economy is dominated by smallholder agriculture due to its highly rural population and lack of industrialization. For several decades, Malawians have participated in the charcoal value chain as a lucrative supplementary livelihood strategy, and growth of Malawi's charcoal economy has accelerated in recent years (Kansanga et al., 2025; Smith et al., 2017, 2019; Sumba et al., 2021; Zulu & Richardson, 2013). Between 18% and 29% of Malawi's land area is forested (Skole et al., 2021), demonstrating a large availability of woody biomass resources to support the woodfuel sector. Malawi's three regions (North, Centre, and South), though all dominated by rural populations, are heterogenous across demographic, economic, and environmental dimensions. Malawi is therefore highly representative of rural regions throughout Eastern and Southern Africa. In many ways, it is a bellwether for other low-income and predominantly rural countries in SSA that are on similar economic, demographic, and environmental change trajectories.

While many studies have characterized charcoal consumption in urban areas and documented charcoal production as a livelihood strategy among rural populations, no studies to date have examined charcoal use among rural populations using nationally representative data over a long timescale. Using publicly available, nationally-representative, and geospatially explicit household survey and environmental change data from Malawi, we address this gap. We describe and test the hypothesis that the emergence of the "charcoal economy," first driven by demand for charcoal among urban populations, has led to a growing reliance on charcoal production and consumption among rural populations. In our analysis, we show how several interrelated transitions (including demographic, economic, and land use) have catalyzed livelihoods changes among rural populations, resulting in both increased participation in the charcoal sector and uptake of charcoal for cooking. We also explore the impact of the woodfuel transition on forest resources, identifying that renewed attention from scholars and policymakers should be paid to woodfuel extraction in sub-Saharan Africa as populations undergo the woodfuel transition.

We build on analysis by Jagger & Perez-Heydrich (2016) who explored the relationship between land use and land cover change and household cooking fuel choices in Malawi and found that deforestation is associated with increased reliance on purchased woodfuels. Our paper reveals new insights by broadening the scope of their empirical work to consider major demographic and economic trends, extending the period under consideration by a decade, and using finer-resolution geospatial measures of forest loss (30-meter rather than 500-meter). Our analysis covers the years 2004 to 2020, when charcoal's rise in popularity was dramatic.

2. Theoretical framework: Interrelated transitions linked to charcoal uptake

Our primary research question is "How are household energy transitions influenced by concurrent demographic, economic, and land use changes?" We take a livelihoods perspective to investigate rural household fuel use as determined by resources available to the household: land, time, labor, and wealth. These resources, in turn, are influenced by meso- and macro-level phenomena, including demographic, economic, and land use transitions. We explore how the following factors, which measure proximate and underlying factors related to these transitions, influence rural households to purchase and use charcoal over firewood: demographic factors (increased number of households, population density, and changes to household composition), economic changes (diversifying livelihood strategies, market access, increased charcoal production, and household wealth accumulation), and shifting land resources (availability of cropland and proximity to forests) (Fig. 1). These transitions are concurrent and interrelated in rural, agrarian societies in SSA, and we hypothesize that observed changes have catalyzed the growth of the charcoal economy and increased charcoal consumption among rural populations. Below, we briefly review the state of knowledge about these transitions and their links to household energy dynamics in SSA.

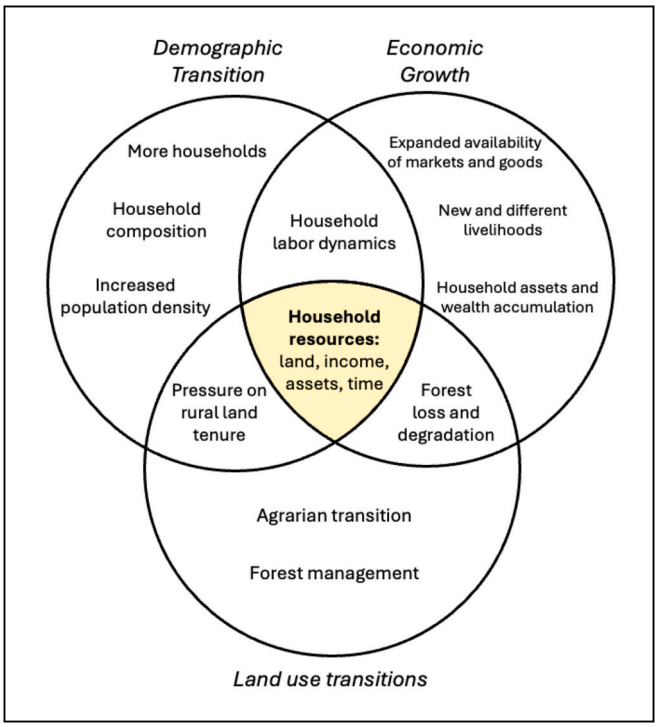


Fig. 1. Theoretical framework: interrelated transitions influence household resources. From a livelihoods perspective, underlying and proximate factors influence rural household resources, which in turn influence household fuel choice.

2.1. Demographic transitions and household energy demand

Population growth, urbanization, rural to urban migration, and increases in the number of households are all dimensions of demographic change theorized and observed to influence energy transitions (Knight & Rosa, 2012; Wu et al., 2021). Urbanization shifts the population traditionally reliant on subsistence agriculture to urban settings with different economic activities, enabling households to have more cash income available for household expenditures, such as food and energy (Gutu Sakketa, 2023). Charcoal, in high demand by urban populations for decades, is produced in distant rural areas, then transported to urban centers, passing through rural markets along trade routes (Kambewa et al., 2007; Zulu & Richardson, 2013). Consequently, growth in urban demand for charcoal increases its presence in rural energy marketplaces and trading centers and along transportation routes.

In many sub-Saharan African populations, the number of households has grown even as average household size has declined, indicating that people increasingly form new, smaller households rather than remaining within larger existing ones (United Nations, 2019). Because smaller households consume greater volumes of fuelwood per capita on average, a shift toward smaller households implies rising total fuelwood consumption (Knight & Rosa, 2012). From a livelihoods perspective, smaller households also have implications for fuel choices. Declines in household sizes change the structure of households and shift the availability of financial and labor resources, which are directly relevant to decisions to purchase or collect woodfuels (de Sherbinin et al., 2008). As a result, households may increase their reliance on cash generating livelihoods instead of subsistence agriculture. Cash generation allows for greater commerce and trading activity (Alobo Loison, 2015; Ellis & Freeman, 2004), including the purchase of household cooking fuels. Meanwhile, reliance on non-agricultural livelihoods introduces constraints on time and labor resources for household tasks such as fuel gathering, which is typically carried out by women and girls (Das et al., 2023; Wassie et al., 2021; Zulu & Richardson, 2013).

2.2. Economic changes and household energy demand

Economic change in rural SSA populations is characterized by diversification of livelihood strategies, growth in accumulated assets, and expanded market access (Habib et al., 2023). An important consequence of growth in the number of households in a population is decreased availability of rural land for farming. Pressure on rural smallholder agriculture is well documented in Malawi and especially affects younger generations, who have less access to farmland and are planting smaller areas than previous generations (Lindsjö et al., 2021). This pressure is further amplified by the agrarian transition in Malawi, represented by an increase in medium- and large-holder farms in the past two decades (Jayne et al., 2019, 2022). In Malawi, approximately 10% of customary land (land maintained by traditional governance, rather than state or private ownership) was acquired by medium- and large-holder farms between 2005 and 2015 alone (Anseeuw et al., 2016). As a result, a small number of relatively well-off farmers occupy large land holdings, and many young farmers face land constraints (Anseeuw et al., 2016; Jayne et al., 2022; Lindsjö et al., 2021).

Disruption in smallholder agriculture as a reliable livelihood strategy has forced rural youth to diversify their livelihood strategies in their communities, primarily through piecework and wage labor (Chinsinga & Chasukwa, 2018). Households that diversify their livelihoods through both farm and non-farming activities achieve greater wealth (Ellis & Freeman, 2004; Habib et al., 2023). Therefore, wealth and asset accumulation are signs that rural households are engaging in livelihood diversification (Ayanwale et al., 2023; Brockington & NOE, 2021). In rural Malawi, accumulated wealth proxied by asset values is a more reliable measure of household economic wellbeing than cash income because the latter is difficult to measure in contexts where livelihoods are made up of diverse components and with seasonal variation in income flows (Howe et al., 2011; Jagger et al., 2012).

Forest-based livelihoods are important alternatives to agriculture, particularly when high-value forest products can be accessed and sold by nearby rural populations (Angelsen et al., 2014; Jagger et al., 2022). As a result, the role of marketplaces and trading centers are important to rural energy access for both sales and consumption. While some smallholders have turned to tree planting and harvesting for charcoal production as a supplementary livelihood strategy (Iiyama et al., 2017), the dominant strategy is to access woody biomass from nearby protected forest areas for charcoal production, tobacco curing, and brickmaking (Kambewa et al., 2007; Khundi et al., 2011; Woollen et al., 2016; Zulu & Richardson, 2013). Therefore, spatial and geographical distributions of populations and land cover change must be carefully considered when exploring how livelihood diversification, land, and energy transitions are linked.

2.3. Land, woodfuel dynamics, and energy demand

Historically, studies of woodfuels have analyzed firewood and charcoal together, obscuring the relative contribution of each fuel separately to deforestation and degradation (Mwampamba et al., 2013). However, it is estimated that informal charcoal production in SSA negatively impacts the environment as much as, if not more than, firewood production (Bailis et al., 2015; Hosonuma et al., 2012). This is because the majority of charcoal production is carried out using inefficient kilns, resulting in greater wood consumed per calorific unit (Bailis et al., 2015; Chidumayo & Gumbo, 2013). Additionally, charcoal is produced from large branches and trunks, leading to tree cutting for wood extraction. Urban charcoal consumption has been connected directly to forest degradation in Malawi's neighboring countries of Mozambique and Zambia, whose populations are experiencing similar energy transition dynamics (Ryan et al., 2014; Sedano et al., 2020, 2021, 2022; Woollen et al., 2016). These studies illustrate that charcoal is produced in rural regions for sale in urban centers, highlighting that production and consumption in the charcoal economy are

geographically decoupled but linked through complex supply chains (Jagger & Shively, 2015; Sedano et al., 2020, 2021).

From the perspective of a population-environment framework, charcoal and firewood consumption differ because charcoal is typically purchased at trading centers or markets, while households collect firewood for their own use. Traditionally, rural households have obtained firewood from nearby forests and woodlands or residual biomass from smallholder land clearing for no monetary cost and relying on household labor (Arnold et al., 2006; Chidumayo & Gumbo, 2013; Jagger & Shively, 2014). Household firewood gathering has a limited impact on proximal forests (Angelsen et al., 2014; Arnold et al., 2006; Bailis et al., 2015; Masera et al., 2015), though deforestation increases the time that households spend collecting firewood (Jagger & Perez-Heydrich, 2016; Jagger & Shively, 2014). On communal lands, deforestation and forest degradation are primarily due to agriculture extensification and woodfuel extraction for tobacco curing and brick making, and there is little evidence that woodfuel production for household cooking incentivizes smallholder clearing (Chidumayo & Gumbo, 2013; Elnagheeb & Bromley, 1994; Hosonuma et al., 2012; Iiyama et al., 2017; Ngwira & Watanabe, 2019).

In Malawi, dense and mature forests, particularly those in government designated national parks, forest reserves, and game reserves, have long been major sites of informal woodfuel production (Chibwana et al., 2013; Jumbe & Angelsen, 2011; Kambewa et al., 2007). Recent studies of protected areas in Malawi have examined the extent and causes of land cover change in Kasungu National Park (Kpienbaareh et al., 2022), Liwonde, Kaning'ina, Thambani (M. F. K. Gondwe et al., 2022), Thuma (Phiri & Nyirenda, 2022), and Dzalanyama forest reserves (Munthali & Murayama, 2011, 2015). Studies focused on Dzalanyama, Liwonde, Kaning'ina, Thambani, and Thuma forest reserves noted that reserves were affected by high rates of woody biomass extraction, though it was not possible to differentiate the motivation for extraction of biomass resources between household woodfuels (firewood and charcoal), tobacco curing, or brick making. Charcoal production also occurs outside of protected areas on customary or privately governed lands in Malawi. However, attribution of forest loss to a particular cause is often not possible using remotely sensed data alone, and other data on charcoal production activities are limited or non-existent due to the informal nature of the sector.

2.4. Hypothesized links between demographic, economic, and land use factors and rural energy transitions

We argue that the dynamics of the charcoal economy are self-reinforcing and amplifying production and consumption in rural populations. We hypothesize that smaller households, headed by younger individuals, and without access to land for farming are growing in number and most likely to use charcoal. Without sufficient agricultural land, these households must diversify their livelihood strategies through cash-generating activities. Such households therefore have resources to purchase charcoal and, because they are smaller, often lack labor resources to gather firewood. Additionally, we hypothesize that rural charcoal consumption is geographically determined. Specifically, households with proximity to markets, trade routes, or sites of charcoal production are most likely to use charcoal, while households with access to abundant woody biomass are most likely to continue to rely on firewood.

3. Methods and data

To study the spatiotemporal dynamics of rural energy transitions, we leverage four waves of nationally representative, cross-sectional survey data from Malawi's Integrated Household Survey to analyze household fuel use in Malawi between 2004 and 2020. Using global, publicly available, high-resolution geospatial Global Forest Cover Data, we integrate household survey data and geospatially and temporally explicit measures of environmental change to investigate relationships between concurrent energy, demographic, economic, and land use transitions. We first characterize changes among Malawi's rural population over the study period by analyzing trends in fuel use, demographic, economic, and land cover indicators. Then, we fit pooled, multi-level, multinomial logistic regression models to evaluate whether hypothesized factors are associated with rural household fuel choices. Finally, we characterize geographic patterns of forest cover loss to demonstrate the environmental impact of the woodfuel transition in Malawi. Our replication package, including datasets and code scripts for all analyses, is available on GitHub.

3.1. Datasets

3.1.1. Survey data

We analyzed the Integrated Household Survey (IHS), a publicly available dataset collected by the Government of Malawi's National Statistical Office and The World Bank (National Statistical Office., 2012; National Statistical Office, 2020; National Statistical Office, 2015; National Statistical Office, 2017). Malawi IHS Waves II-V[1] are nationally representative, cross-sectional surveys of households and enumeration areas (EAs) from which households were selected. The IHS selected households from Malawi's census using a two-stage clustered sampling design. First, a random stratified sample of EAs (the smallest administrative area defined by Malawi's census, which may represent part of a village, one or several villages, or part of an urban area) were selected from district-specific urban–rural strata; in the second stage, a random systematic sample of households were selected from EAs. Stratified EA sampling ensured an efficient sample and allowed analysis to be decomposed by urban–rural strata. Urban-rural strata were assigned by Malawi's census and leveraged by the IHS. Further details about the design of the IHS and its microdata are available for public use at the World Bank's Microdata Library.

In total, 47,208 (39,203 rural; 8,005 urban) households across a total of 2,815 EAs[2] were surveyed across four waves. By wave, the surveyed households and EAs totaled: 11,280 households across 564 EAs in IHS2; 12,271 households across 768 EAs in IHS3; 12,225 households across 768 EAs in IHS4; and 11,402 households across 715 EAs in IHS5. The IHS included geographic coordinates for each EA offset up to five kilometers within the EA's originating Traditional Authority (the administrative unit in Malawi between district and EA, comparable to a county in the United States).[3] We exclude 564 households (1.1%) from our analysis due to missing geographic coordinates or EA information. Therefore, our pooled analytical dataset totaled 46,644 households.

We extracted variables relevant to our analysis from both household and EA surveys; Table A1.

in the Appendix and replication scripts include details about variable harmonization across IHS waves. Primary cooking fuel (collected firewood, purchased firewood, charcoal, electricity, other) and supplementary charcoal use (when a household reported buying charcoal in the past 7 days but used a different primary fuel) were obtained from household surveys. To measure firewood collection behavior and access, we calculated households' total time spent collecting firewood in the 24 hours preceding the survey divided by the number of members, restricted to individuals aged five and older. Other household measures of demographic composition and economic wellbeing included: age and sex of household head and all other members, acres of rainfed agriculture planted in the preceding year, and total value of durable assets

[1] Data from IHS I are not included in analysis because geographic coordinate data are not available.

[2] We exclude data from Likoma District, a small island in Lake Malawi, because the district was not included in all four waves of the IHS.

[3] A random 1% subset of EAs are offset up to ten kilometers to further ensure participants' anonymity.

(inflation-adjusted Malawi Kwacha [MWK]) (World Bank, 2025).

EA surveys provided information about EA amenities and characteristics reported by local leaders and key informants. From EA surveys, we obtained information about whether the EA hosts a daily or weekly market (yes/no), distance to nearest major road (kilometers), and whether the woodfuel value chain was among the top three livelihood strategies for the EA's population (yes/no). Market variables and distance to major roads approximated market and trade route access. Community reliance on the woodfuel value chain (e.g., production, transport, or trade of firewood or charcoal) provided information about participation in the woodfuel sector at the community level, which was an important measure in our analysis, as woodfuel sector participation was not available at the household level.

3.1.2. Geospatial data

We measured forest cover and forest cover loss for local areas around sampled EAs using the Global Forest Cover Data v1.9 (GFCD) (Hansen et al., 2013). GFCD are a publicly available dataset derived from Landsat data and characterize global tree canopy cover at a 30-meter resolution; we downloaded the data from the public Global Forest Change Data Download repository. We used baseline tree canopy cover in 2000 and annual tree cover loss variables from the dataset. In areas surrounding the EA, forest cover approximated availability of woody biomass for household firewood collection, and forest loss approximated woody biomass loss and potential residual biomass available to households. The GFCD defined loss as significant disturbance and/or complete loss of tree cover. Therefore, the measure of forest loss in our study did not include low grade forest degradation nor did it necessarily indicate deforestation (Hansen et al., 2013). Spatially explicit and temporally disaggregated forest regrowth data were not available for the entire study period from the GFCD, and inability to account for regrowth is a limitation of our analysis.

We calculated year 2000 forest cover and temporally precise forest cover loss within a 10-kilometer radius buffer around each sampled EA. A 10-kilometer radius accounted for spatial imprecision of IHS geographic coordinates due to spatial offsets (randomly assigned distances up to five kilometers); an additional five kilometers to capture the maximum distance that households generally travel to gather firewood (Jagger & Jumbe, 2016). To calculate baseline forest cover at year 2000 for each EA, we summed the area of 30% or greater tree canopy cover (the threshold recommended by the GFCD to represent forest cover) within the buffer area (Potapov et al., 2021).[4] We calculated forest loss as the average annual rate of loss (square kilometers per year in the buffer area) between the previous survey wave and final year of the given wave. To encode forest cover loss, the GFCD assigns each pixel a value from 2001 to 2020, corresponding to the first year of major forest loss or degradation. If no forest cover loss was detected between 2001 and 2020, the pixel was assigned a value of 0. To calculate forest cover loss rates, we summed the total area of forest cover loss in the relevant wave-years divided by number of years. Years for each survey wave were: 2001–05 for IHS2; 2006–11 for IHS3; 2012–17 for IHS4; 2018–20 for IHS5.

We analyzed patterns of forest cover loss in protected areas and explored how household proximity to these areas may be associated with fuel use. Shapefiles of protected areas were downloaded from the Government of Malawi's Department of Forestry website on March 1, 2023 (Government of Malawi, 2023). This dataset included geographic boundaries for 26 federally managed forest reserves, game reserves, and national parks. Using geospatial intersection, we created an indicator of proximity to any protected area (within 10 km of any n = 26 protected areas, yes/no) for each EA. We created a second indicator of proximity to a subset of protected areas (n = 6), which we call "key protected areas," where > 10% of the forest cover area in 2000 was lost between 2001 and 2020, and no formal woodfuel extraction activities nor major fires have been documented. We infer these key protected areas to be sites of intensive, unsanctioned charcoal production in analysis.

Population density values for EAs were derived from WorldPop gridded population estimates from the years of IHS data waves: 2004, 2010, 2016, and 2019 (WorldPop, 2018). We extracted mean population density values for all pixels within the 10-kilometer buffer area around each EA to capture variation in local population density. We prepared all geographic data in R version 4.3.1 using the "terra", "exactextractr," and "sf" packages (Baston, 2023; Hijmans, 2024; Pebesma, 2018; R Core Team, 2023).

3.2. Descriptive statistics and trends

Using survey weights assigned by the IHS, we generated nationally representative estimates of household fuel use. Estimates were representative to urban/rural strata at the national level (Lumley, 2010). We created stacked bar plots to show trends in primary fuel use across the national population, disaggregated by urban/rural designation and wave. To explore the growth of charcoal consumption in absolute terms, we estimated and plotted the total number of rural and urban households using charcoal as primary and supplementary fuel over time (Wickham, 2016).

We explored national trends in rural households' and EAs' characteristics during the study period. A total of 38,847 rural households were included in descriptive analysis. We calculated and reported descriptive statistics for household and EA characteristics disaggregated by IHS wave. To assess linear trends in the variables across the study period, we fit univariate regression models for household and EA variables. We regressed each variable on the first year of the respective IHS survey wave (2004 for IHS2; 2010 for IHS3; 2016 for IHS4; 2019 for IHS5) so that coefficients represented average annual linear changes in each variable. We implemented robustness checks in the linear trend analysis by removing outliers and refitting the univariate regressions. Outliers were identified using Tukey's fences: values falling more than 1.5 times the interquartile range below the first quartile or above the third quartile of the pooled observations.

3.3. Multi-level, multinomial logistic regression modeling

To test whether hypothesized factors were associated with rural households' fuel choices, we fit multi-level, multinomial logistic regression models at the household level using the pooled dataset. A random intercept for EA was included to account for the clustered sampling design, and a survey-wave fixed effect adjusted for temporal trends in fuel use unexplained by predictor variables. As our goal was to identify conditional associations between hypothesized factors and fuel choice rather than to produce population-representative estimates, we did not apply survey sampling weights in the models. Each model included the same predictor variables so that relative effect sizes and significance values were comparable among fuel use outcomes. A total of 38,668 household observations were included in modeling analysis; we excluded 178 observations (0.5%) due to missing information about primary fuel (n = 32), distance to nearest major road (n = 136), or age of household head (n = 10). We separately modeled household primary fuel (referent group = collected firewood) and supplementary charcoal use (referent group = no charcoal use). We reported exponentiated coefficients (adjusted odds-ratios, ORs), 95% confidence intervals (95% CIs), and significance values. We fit models using the "mclogit" package in R (Elff, 2022).

We included the following predictor and control variables in both models: per capita value of household assets (MWK adjusted for inflation; log-10 transformed), area of rainfed agriculture cultivated by the household (acres), age of the household head (years; mean centered on

[4] Forest extent and loss area values are based on reprojected GFCD data in the WGS 84/UTM zone 36S (EPSG:32736) projected coordinate system, which is appropriate for Malawi.

43.4, and scaled); percentage of the household members that were female (10% increase in female representation; centered on 50%); household size (1 person; mean centered on 4.5); availability of daily or weekly market in EA (1/0); EA within 10 km of a key protected area (1/0); EA distance to major road (kilometers, log-2 transformed); year 2000 forest in the EA buffer area (square kilometers); rate of recent forest cover loss in the EA buffer area (square kilometers/year); population density (people/kilometer; log-2 transformed); fixed effect for the national region in which the household resides (referent group=Centre); and fixed effect for the wave in which the household was surveyed (referent group = IHS2). Predictors were transformed or scaled to improve model fit and interpretation. To avoid log(0), a positive constant of 0.001 was added to each log-transformed variable.

We tested for model misspecification to ensure that unspecified geographic effects, such as preference diffusion through geographic proximity, were not influencing modeled associations. To do so, we analyzed the spatial autocorrelation of EA random effects; this subanalysis focused on random effects from the model of primary fuel use. Spatial clustering of the random effects would have indicated potential bias due to the omission of geographic factors linked to either the explanatory variables or the modeled outcome (Getis, 2010). Global Moran's I was calculated to assess the degree of global spatial autocorrelation among random effects for each wave separately; a distance weighting scheme was used so that each EA included at least one neighbor.[5] Additionally, local indicator of spatial autocorrelation (LISA) analysis was conducted to identify geographically explicit location of any spatial autocorrelation among residuals. We observed minimal indication of global and local spatial autocorrelation, and maps of LISA results were included in the Appendix. The global and local spatial autocorrelation analysis was conducted using the "spdep" package in R (Pebesma & Bivand, 2023).

3.4. Geographic analysis of forest cover loss

We explored geographic and temporal patterns of forest cover loss in national protected areas to identify areas affected by informal woodfuel extraction. Protected areas in Malawi are designated as one of three categories: Forest Reserves, National Parks, or Game Reserves. These areas are administered by the Department of Forestry through the Ministry of Natural Resources. Aside from the operation of timber plantations, construction of new administrative centers, and other sanctioned agricultural programs within these areas, the forests inside protected areas are not allowed to be cleared. Therefore, we attribute forest loss to either unsanctioned woodfuel extraction, or less commonly, forest fires. For each protected area, we calculated baseline forest extent area for the year 2000, and the total area of forest loss by each wave in square kilometers. We mapped forest loss patterns in the subset of protected areas that were very likely sites of unsanctioned woodfuel extraction, which we call "key protected areas" in our analysis (where > 10% of the forest cover area in 2000 was lost between 2001 and 2020, and no formal woodfuel extraction activities nor major fires have been documented).

4. Results

4.1. Growth of charcoal consumption in Malawi

Charcoal consumption in Malawi grew substantially between 2004 and 2020, in both absolute number and relative share of households. By 2020, more rural households used charcoal than purchased firewood as their primary fuel. Fig. 2 shows the share of households that used one of five primary fuels for cooking, disaggregated by survey wave and urban/rural designation. The percentage of rural households who used charcoal as their primary fuel increased from 1.2% [95% CI, 1.0–1.4] in 2004 to 7.4% [6.9–8.1%] in 2020. This corresponded to a nine-fold increase in the number of rural households in Malawi using charcoal as their primary cooking fuel, from 28,000 households in 2004 to 254,000 in 2020, reflecting both rural population growth and expansion of charcoal's popularity overall. Notably, this increase in primary charcoal use corresponded to a decrease in purchased firewood as primary fuel among rural households, from 13.2% (315,000 households) to 6.4% (219,000 households) over the study period.

Among urban households, charcoal was the most popular primary fuel across all four waves and increased in popularity from 50.4% [47.8–52.3%] in 2004 to 74.9% [72.6–77.0%] in 2020. Considering growth in absolute terms, the number of urban households in Malawi nearly doubled from 353,000 to 652,000 between 2004 and 2020. In the same period, the number of urban households using charcoal as the primary cooking fuel more than doubled from 178,000 to 448,000. As with rural households, we observed a parallel, and even more substantial decrease in urban households using purchased firewood as their primary cooking fuel, from 23.5% (83,000 households) to 8.1% (53,000 households).

Finally, we note that electricity used for cooking among urban households was highly variable throughout the study period, declining from 11.5% [9.8–13.4%] to 6.1% [4.8–7.9%] (Fig. 2). Because the factors that determine household use of electricity for cooking are theoretically different than those determining woodfuel choices (charcoal and firewood), discussion of electricity use among urban households were beyond the scope of our analysis. We address this point briefly in the discussion as it relates to charcoal use.

Between 2004 and 2020, the number of households using charcoal as either primary or supplementary fuel more than tripled nationally, from 321,000 (out of 2.72 million households nationally) to 1.02 million (out of 4.07 million households nationally) (Fig. 3). This growth represented a doubling in the percentage of households using charcoal in any form, from 11.8% to 25.0%. Rural households using any charcoal (either primary or supplementary) grew from 91,000 to 458,000, and the number of urban households using any charcoal grew from 230,000 to 559,000. Estimate values represented in Figs. 2 and 3 are reported in tabular form in the Appendix (Table A2).

4.2. Household and environmental changes from 2004 to 2020

We observed increased charcoal use (both as a primary and supplementary cooking fuel) and decreased purchased firewood use during the study period (Table 1). However, we observed no change in the proportion of households using collected firewood, electricity, or other biomass (crop residues and dung) as their primary fuels. Additionally, there was no apparent change in the average time spent by household members collecting firewood. Per capita wealth fluctuated non-linearly. This could be due to fluctuations in Malawi's economic trends, and the linear coefficient was therefore not statistically significant ($\beta = 831.00$; $P > 0.05$). The median size of rainfed cropland planted decreased from 1.8 acres to 1.0 acres over the study period, and a linear decreasing trend was statistically significant ($\beta = -0.03$; $P < 0.01$). The percentage of households headed by females grew steadily over the study period, from 24% in 2004–05 to 32% in 2019–20 ($\beta = 0.03$; $P < 0.01$). The linear coefficients for average household sizes, sex composition of the households, and average age of household heads were very marginal (though statistically significant) and therefore provide little evidence of substantial change nationally.

We did not observe a statistically significant change in EAs' market access, as approximated by proportion of EAs with daily or weekly market or distances to major roads (Table 2). However, the number of EAs that reported reliance on the woodfuel (firewood and charcoal)

[5] Spatial autocorrelation indices of random effects are conducted for each IHS wave separately, and therefore the neighbor's matrices have different distance threshold values: 49.2 km for IHS2; 22.9 km for IHS3; 23.8 km for IHS4; 25.0 km for IHS5.

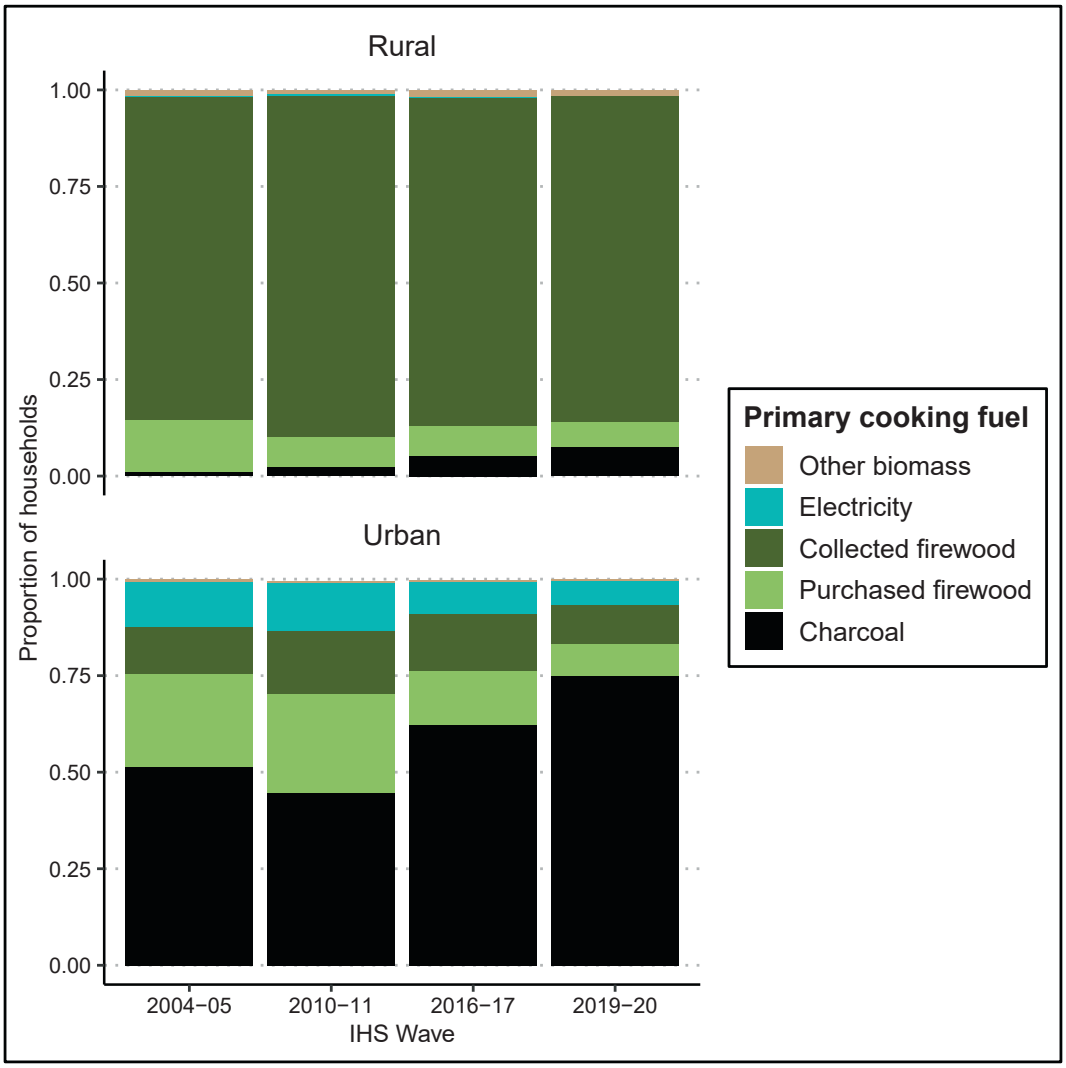


Fig. 2. Primary cooking fuel prevalence in Malawi, 2004–2020. Household primary fuel use disaggregated by urban–rural designation. Weighted national estimates derived from IHS II-V.

sector nearly doubled, from 11% to 19% ($\beta = 0.05$; $P < 0.01$). Additionally, rates of forest cover loss in areas surrounding the EAs increased over the study period, from an average of 0.12 to 0.32 square kilometers per year. The coefficient for forest loss linear trend was statistically significant ($\beta = 0.02$; $P < 0.01$), though not robust to outlier removal; this reflected the highly skewed distribution of forest loss values in the dataset. As expected, there was no difference between year 2000 forest cover values or number of EAs within 10 km of protected areas over time, which indicated sufficiently random geographic coverage across survey waves.

4.3. Factors associated with rural woodfuel choices

Rural households with higher per capita value of assets, headed by younger individuals, living in high population density EAs with daily or weekly markets, close to major roads, and within 10 km of key protected areas had greater odds of using charcoal and purchased firewood, compared to collected firewood (Fig. 4). The direction and statistical significance for these effects were consistent and similar across both outcomes. However, the effect sizes for purchased firewood were relatively weaker and were significantly different at a $P < 0.05$ level, confirmed by contrasts between charcoal and purchased firewood. Proximity to key protected areas was one exception, with similar magnitude and statistical significance for both charcoal and purchased firewood.

There were four important differences between factors associated with using charcoal or purchased firewood. First, for each additional acre of rainfed agriculture planted by the household, the odds of using charcoal as the primary fuel decreased 9% (adjusted OR, 0.91; 95% CI, 0.86–0.96) when compared to collected firewood; no association between area planted and purchased firewood was observed. Second, for each additional household member, the odds of using purchased firewood increased 5% (adjusted OR, 1.05; 95% CI, 1.03–1.07), but the odds of using charcoal decreased 7% (adjusted OR, 0.93; 95% CI, 0.89–0.96) compared to collected firewood. Third, female headed households had lower odds of using charcoal as their primary cooking fuel choice compared to collected firewood (OR, 0.82; 95% CI, 0.69–0.98); however, sex of household head was not associated with charcoal use when compared to purchased firewood. Finally, we observed no association between living in an EA that relies on the woodfuel value chain (producing, transporting, and selling firewood and charcoal) and charcoal use when compared to collected firewood. However, a household's odds of using charcoal were 1.43 times the odds of using purchased firewood in communities reliant on the woodfuel sector (95% CI, 1.06–1.91; contrasts calculated from model output, not shown in Fig. 4).

Younger household head, higher per capita wealth, and living in an EA with market access, higher population density, and nearer to major

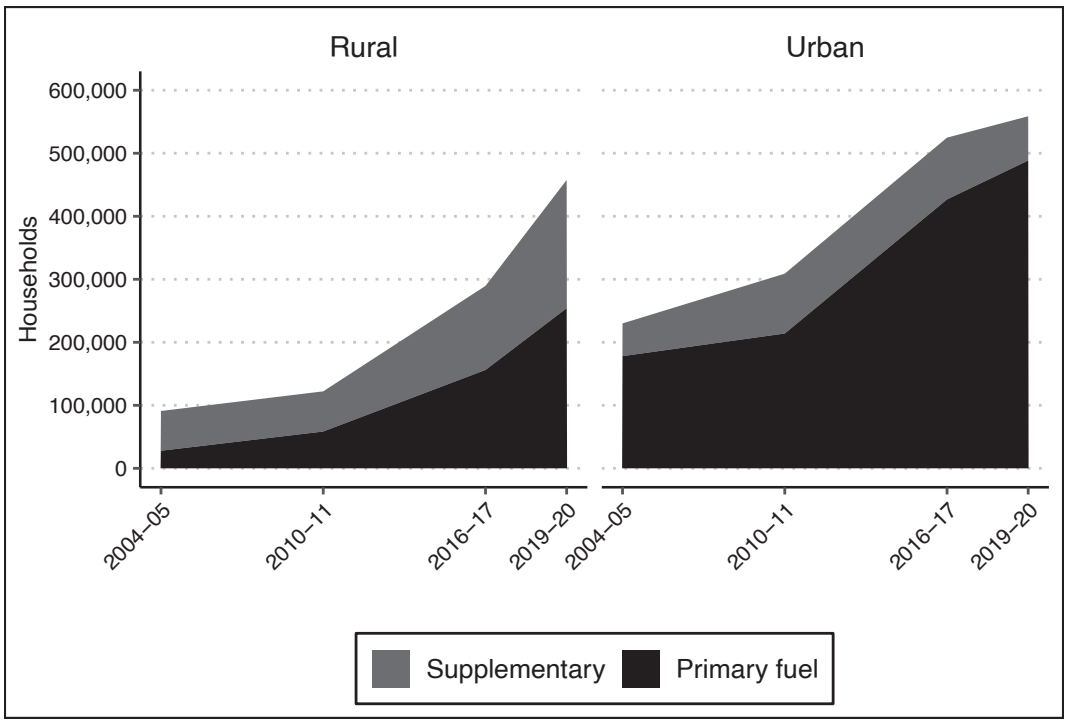


Fig. 3. Households using charcoal as primary or supplementary fuel in Malawi, 2004–2020. Numbers of households disaggregated by urban/rural strata. Weighted national estimates derived from IHS II-V.

Table 1
Sample rural household characteristics in Malawi, 2004–2020.

	IHS2 (2004–05)	IHS3 (2010–11)	IHS4 (2016–17)	IHS5 (2019–20)	β: Linear coefficient[a]
	(N = 9840)	(N = 10038)	(N = 9763)	(N = 9206)	
Primary cooking fuel					
Charcoal	112 (1.1%)	211 (2.1%)	496 (5.1%)	730 (7.9%)	0.14*** [b]
Purchased firewood	1278 (13.0%)	694 (6.9%)	667 (6.8%)	556 (6.0%)	−0.05*** [b]
Collected firewood	8283 (84.2%)	8973 (89.4%)	8386 (85.9%)	7770 (84.4%)	0.00[b]
Electricity	35 (0.4%)	43 (0.4%)	25 (0.3%)	34 (0.4%)	0.00[b]
Other biomass	120 (1.2%)	105 (1.0%)	182 (1.9%)	114 (1.2%)	0.02* [b]
Charcoal use					
Primary fuel	112 (1.1%)	211 (2.1%)	496 (5.1%)	730 (7.9%)	0.14*** [b]
Supplementary	249 (2.5%)	229 (2.3%)	405 (4.1%)	594 (6.5%)	0.07*** [b]
None	9467 (96.2%)	9586 (95.5%)	8855 (90.7%)	7880 (85.6%)	−0.11*** [b]
Time collecting firewood (daily hours/household member)					
Mean (SD)	0.15 (0.38)	0.18 (0.32)	0.16 (0.31)	0.17 (0.37)	0.00
Household asset values (2010 MWK/capita)					
Median [Q1, Q3]	1160 [461, 2754]	1325 [440, 3162]	1554 [553, 3797]	1285 [431, 3466]	831.0
[Min, Max]	[0, 2105419]	[0, 819900]	[0, 196748906]	[0, 2565112]	
Rainfed agriculture land (acres)					
Median [Q1, Q3]	1.8 [1.0, 3.0]	1.5 [1.0, 2.3]	1.0 [0.5, 2.0]	1.0 [0.5, 2.0]	−0.03***
[Min, Max]	[0.0, 9800.0]	[0.0, 30.0]	[0.0, 30.0]	[0.0, 159.9]	
Age of household head (y)					
Mean (SD)	43.2 (16.6)	42.9 (16.7)	43.8 (16.7)	43.9 (16.6)	0.05***
Sex of household head					
Female	2362 (24.0%)	2529 (25.2%)	2919 (29.9%)	2924 (31.8%)	0.03*** [b]
Male (Ref.)	7478 (76.0%)	7507 (74.8%)	6844 (70.1%)	6282 (68.2%)	
Household size					
Mean (SD)	4.6 (2.3)	4.6 (2.2)	4.3 (2.0)	4.5 (2.1)	−0.01***
Median [Min, Max]	4 [1, 27]	4 [1, 19]	4 [1, 17]	4 [1, 22]	
% female household members					
Mean (SD)	49.6 (22.7)	50.3 (22.6)	51.6 (23.4)	51.4 (23.4)	0.13***

[a] Coefficients from univariate linear regression models predicted by first year of given survey wave (2004, 2010, 2016, or 2019). Stars indicate statistical significance: *P-values: < *0.05, < **0.01, ***< 0.001.*

[b] Binary logistic regression coefficient.

roads increased the odds of using charcoal as both primary and supplementary fuel (Fig. 5). However, we also observed some key differences between primary and supplementary charcoal use. A one-acre increase of the area of cropland cultivated by the household decreased the odds of using charcoal as the primary fuel substantially (OR, 0.71; 95% CI, 0.67–0.76); however, there was no association between

Table 2
Characteristics of surveyed rural Enumeration Areas (EAs) and their local environments, 2004–2020.

	IHS2 (2004–05) (N = 492)	IHS3 (2010–11) (N = 628)	IHS4 (2016–17) (N = 617)	IHS5 (2019–20) (N = 576)	β: Linear coefficient [a]
Daily/weekly market					
No	275 (55.9%)	366 (58.3%)	333 (54.0%)	300 (52.1%)	0.01[b]
Yes	217 (44.1%)	262 (41.7%)	284 (46.0%)	276 (47.9%)	
Distance to major road (km)					
Mean (SD)	19.8 (30.7)	26.8 (75.1)	17.3 (29.1)	18.0 (49.8)	−0.33
Median [Min, Max]	9.00 [0, 400]	11.0 [0, 1300]	9.00 [0, 500]	7.00 [0, 900]	
Woodfuel value chain is a major livelihood					
No	440 (89.4%)	572 (91.1%)	533 (86.4%)	469 (81.4%)	0.05***[b]
Yes	52 (10.6%)	56 (8.9%)	84 (13.6%)	107 (18.6%)	
Within 10 km of protected area (PA) [c]					
Non-key PAs	33 (6.7%)	49 (7.8%)	43 (7.0%)	39 (6.8%)	0.00[b]
Key PA (high forest loss)	110 (22.4%)	124 (19.7%)	132 (21.4%)	130 (22.6%)	0.00[b]
Not near PA	349 (70.9%)	455 (72.5%)	442 (71.6%)	407 (70.7%)	0.00[b]
Local area forest cover in 2000 (km^2) [d]					
Mean (SD)	30.0 (39.9)	32.9 (42.5)	32.2 (41.7)	33.2 (41.2)	0.16
Median [Min, Max]	12.9 [0, 293]	14.4 [0, 260]	13.5 [0, 266]	16.2 [0, 269]	
Local area forest cover loss rate (km^2/yr) [d]					
Mean (SD)	0.12 (0.23)	0.16 (0.35)	0.31 (0.66)	0.33 (0.75)	0.02***[R]
Median [Min, Max]	0.03 [0, 2.31]	0.03 [0, 5.96]	0.05 [0, 5.21]	0.04 [0, 6.58]	
Population density (people/km^2)					
Mean (SD)	196 (162)	213 (192)	245 (216)	267 (264)	0.04***
Median [Min, Max]	167 [8, 1470]	170 [6, 1660]	198 [25, 2000]	208 [11, 2770]	
# Households surveyed					
Median [Min., Max.]	20 [20, 20]	16 [15, 16]	16 [12, 16]	16 [12, 16]	

[a] Coefficients from univariate linear regression models predicted by first year of given survey wave. Stars indicate statistical significance: *p-values: < *0.05, < **0.01, ***< 0.001.*

[b] Binary logistic regression coefficient; outlier removal check was not relevant. [R] Coefficient direction and significance was not robust to removal of outlier values; outliers identified as values less than [Q1 – Interquartile Range] or greater than [Q3 – Interquartile Range].

[c] Protected areas analyzed using official boundaries of 26 areas; key PA's had > 10% of year 2000 forest cover area lost between 2001 and 2020, and no documented formal woodfuel extraction activities nor major fires: Dzalanyama Forest Reserve (FR), Chimaliro FR, Liwonde FR, Mchese FR, Musisi FR, and Lengwe National Park.

[d] Local area is 10-kilometer buffer area around offset EA coordinates given by the IHS.

cultivated acreage and supplementary charcoal use. And, for each additional household member, the odds of supplementary charcoal use increased by 11% (OR, 1.11; 95% CI, 1.08–1.14), but household size was unassociated with primary charcoal use compared to no charcoal use. Finally, a household living within 10 km of a key protected area had 1.62 times the odds of using charcoal as a primary fuel compared to no charcoal use at all (95% CI, 1.33–1.98), but there was no association with supplementary charcoal use. This may have to do with the dynamics of charcoal purchasing and access, which we expand upon in the discussion section.

We implemented a robustness check to assess the association between living within 10 km of a key protected area highly affected by forest cover loss and charcoal use by fitting two alternative models. The first check fit a model that had no indicator for proximity to protected area (Models 1a and 2a); the second included an indicator for proximity to any of Malawi's 26 protected areas for which we had geographic boundary data (Models 1b and 2b). Our primary models reported in Figs. 4 (Model 1) and 5 (Model 2) had improved fit over both alternative models. Additionally, Models 1b and 2b show that living in proximity to any of the protected areas was not associated with any of the fuel use outcomes. All coefficient estimates, significance levels, and fit criteria for primary and alternative models are reported in Tables A3 and A4 in the appendix.

Spatial autocorrelation analysis of EA random intercepts indicated no spatial autocorrelation on a national level (global Moran's I for each wave: IHS2 = 0.03; IHS3 = 0.00; IHS4 = -0.04; IHS5 = 0.03). Lack of spatial autocorrelation suggested that the geographic contextual variables were well specified to model charcoal use in Malawi on the national scale. Exploratory results of local indicators of spatial autocorrelation were reported in the Appendix (Fig. A1)) and show some areas where the model underestimated or overestimated charcoal use.

4.4. Forest loss in protected areas

Analysis of GFCD showed that between 2001 and 2020, 10% or more of the forest cover area was lost in nine of Malawi's 26 protected areas: South Viphaya Forest Reserve (FR), Dzalanyama FR, Mulanje Mountain FR, Lengwe National Park (NP), Liwonde FR, Zomba-Malosa FR, Mchese FR, Chimaliro FR, and Musisi FR (Table 3). Among these areas with high levels of forest cover loss, we excluded South Viphya FR, Mulanje Mountain FR, Zomba-Malosa FR from further analysis, because there were large timber plantations within these reserves, and we were therefore unable to distinguish between loss from formal timber harvesting and informal woodfuel extraction. We identified the remaining six areas (Dzalanyama FR, Liwonde FR, Mchese FR, Chimaliro FR, Musisi FR, Lengwe NP) as key protected areas in analysis (as used in our regression models); Fig. 6 shows the locations of these key protected areas in Malawi.

Across the six key protected areas, a total of 19.5% forested area (178.3 km^2) was lost between 2001 and 2020. Notably, loss between 2018 and 2020 alone accounted for 43% of total loss in these areas during the study period. In Dzalanyama FR, Liwonde FR, and Mchese FR, approximately one-fifth of the forest areas detected in 2000 was lost by 2020. The great majority of loss occurred in each of these after 2012: 89.2% (73.5 km^2) in Dzalanyama; 96.5% (30.6 km^2) in Liwonde; 88.9% (12.5 km^2) in Mchese. Fig. 7 shows the patterns of loss in all key areas, color-coded by the period in which loss was detected.

5. Discussion

Concurrent changes in demographics, household and community economics, and land use are reshaping the livelihoods of rural Malawians, and by extension, their fuel choices. Our analysis demonstrates the rapid growth of charcoal for cooking among Malawi's rural population, the nation's largest demographic constituency. Our results

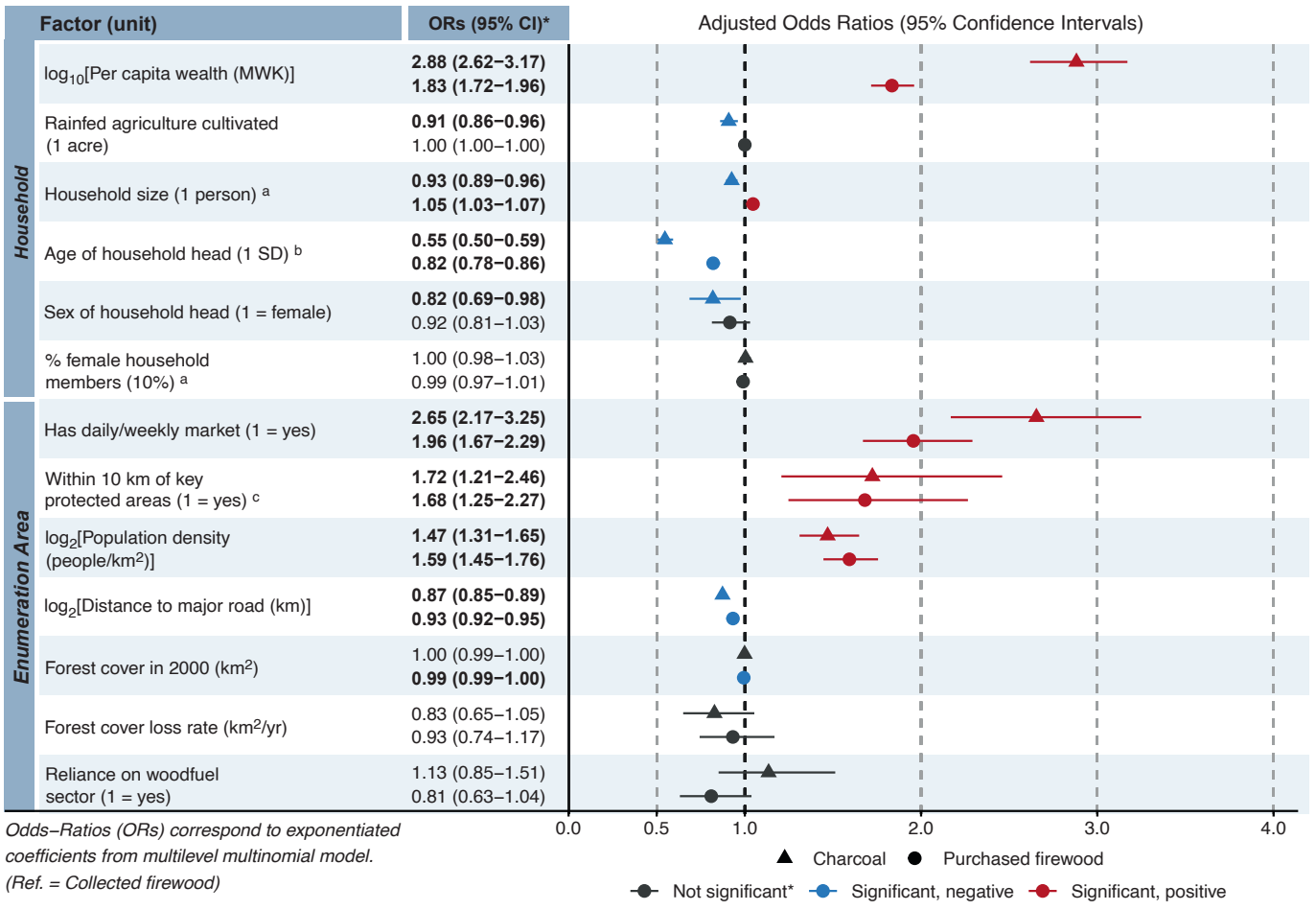


Fig. 4. Factors associated with rural households' primary fuel use in Malawi, 2004–2020. *Bolded ORs indicate statistical significance at the $P < 0.05$ level. All coefficients and P-values for all predictor and control variables are reported in Table A3 in the Appendix. [a] Mean centered variable: % female centered on 50%; household size centered on 4.5. [b] Scaled variable: mean = 43.4 years; s.d. = 16.7 years. [c] Key protected areas are those where > 10% of the forest cover area in 2000 was lost between 2001 and 2020, and no formal woodfuel extraction activities nor major fires have been documented: Dzalanyama Forest Reserve (FR), Chimaliro FR, Liwonde FR, Mchese FR, Musisi FR, and Lengwe National Park.

confirm a "woodfuel transition" from firewood to charcoal in rural households. Rural household energy transitions in SSA receive comparatively less attention than urban energy transitions, as rural households without access to electricity are perceived to rely on freely available firewood to meet cooking energy needs. Our findings illustrate that rural woodfuel choices are changing, highlighting that greater research and policy attention is needed to understand the energy transition dynamics of this large and diverse demographic constituency in SSA, and the woodfuel transition's implications for human well-being and natural resource management.

5.1. *Charcoal is replacing firewood*

Charcoal consumption grew significantly among both rural and urban households in both absolute and relative terms. For urban households with access to centralized grid electricity, Malawi's electricity supply (over 95% generated by hydroelectric power) was highly unreliable during our study period (Government of Malawi, 2020), likely explaining the lack of consistent rates of electricity usage for cooking among the urban population (Fig. 2). Households in Malawi have substituted electricity with charcoal in highly disruptive periods (Smith et al., 2017; Zulu, 2010), demonstrating the structural links between fuel choices and the energy transition. Charcoal therefore represents energy security to urban populations that cannot rely on electricity or alternative fuel supply chains, further entrenching its importance in the energy landscape of SSA (Rose et al., 2022).

Though the proportion of rural households purchasing their primary cooking fuels (rather than using collected firewood) did not change between 2004 and 2020, charcoal appears to have replaced firewood as the purchased woodfuel of choice (Fig. 2 and Table 1). Furthermore, the absolute number of rural households purchasing charcoal grew substantially (Fig. 3), outpacing population growth rates overall. When considering primary and supplementary fuel use combined, the rate of rural charcoal use grew 3.4-fold from 3.9% in 2004 (91,000 households) to 13.4% (458,000 households). Because the informal charcoal sector has greater negative impacts on the environment than informal firewood collection (Bailis, 2009; Bailis et al., 2015), the increase in charcoal's popularity, in parallel with decreased purchased firewood use, raises important questions about sustainable management of woody biomass resources. Furthermore, while rural electrification programs are a priority for Malawi and other countries in SSA, roll out of centralized electricity infrastructure has been slow. In the meantime, households have been adopting solar devices, increasing rates of household electrification by some metrics, though these devices are still not sufficient for cooking (Mahieu et al., 2025; McCord et al., 2026). As such, we can expect that until alternative cooking energy sources are available and accessible to rural households, the growing rural population will contribute to the growing charcoal demand in the region.

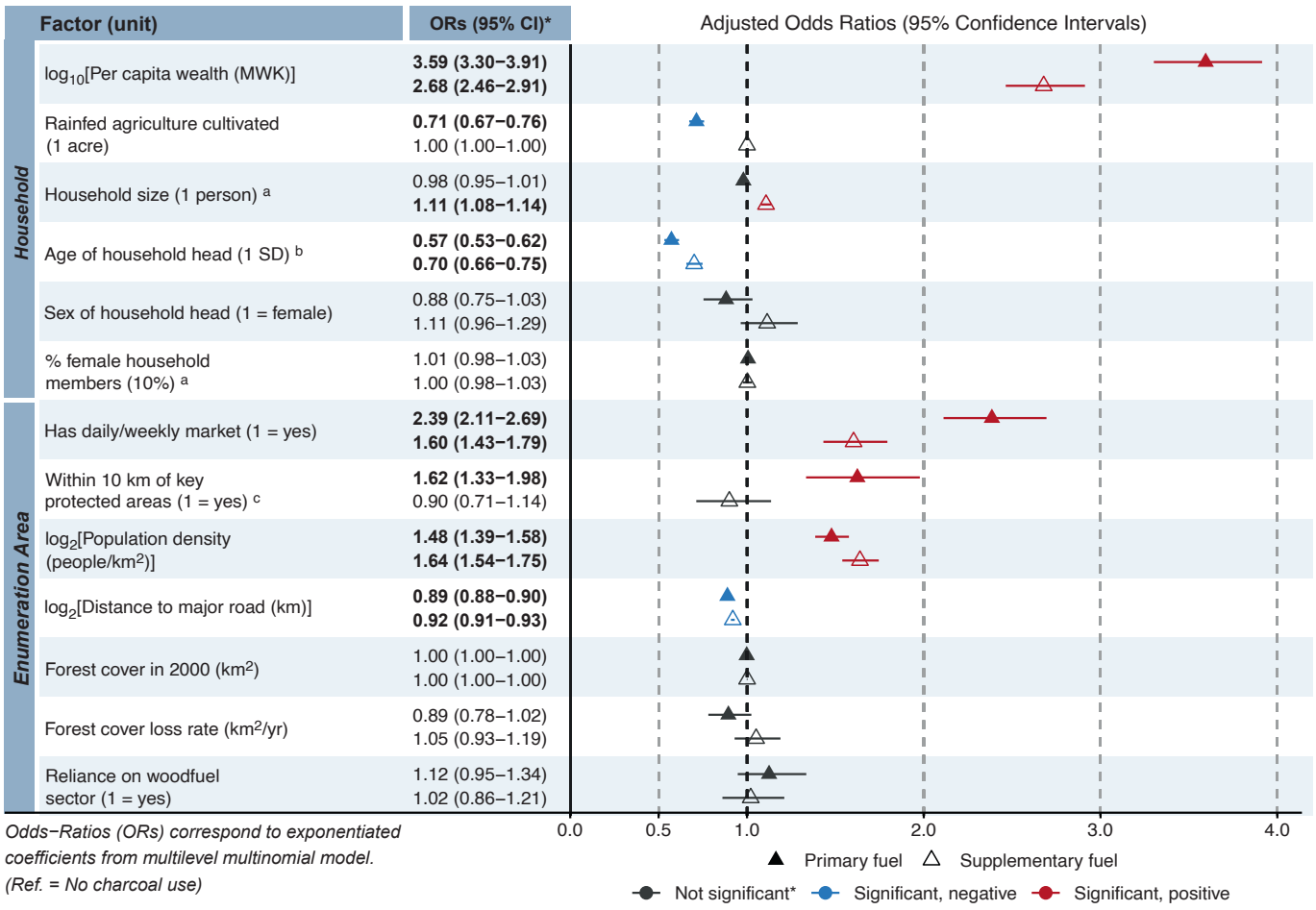


Fig. 5. Factors associated with rural households' charcoal use in Malawi, 2004–2020. *Bolded ORs indicate statistical significance at the $P < 0.05$ level. All coefficients and P-values for all predictor and control variables are reported in Table A4 in the Appendix. [a] Mean centered variable: % female centered on 50%; household size centered on 4.5. [b] Scaled variable: mean = 43.4 years; s.d. = 16.7 years. [c] Key protected areas are those where > 10% of the forest cover area in 2000 was lost between 2001 and 2020, and no formal woodfuel extraction activities nor major fires have been documented: Dzalanyama Forest Reserve (FR), Chimaliro FR, Liwonde FR, Mchese FR, Musisi FR, and Lengwe National Park.

5.2. Why is rural charcoal use growing?

Our analysis demonstrates that charcoal has become widely available in rural areas due to growing production and transportation from rural production sites, through rural markets, to urban centers. A growing rural charcoal user base implies two changes in charcoal's market presence. First, the volume of charcoal for purchase has increased and is no longer bypassing rural markets on its way to urban centers; second, the share of rural households allocating cash to household fuel expenditures has increased. Though we do not observe substantial change in market access across rural communities in our analysis, the number of communities engaged in the woodfuel sector nearly doubles between 2004 and 2020 (Table 2). Without expanded market access, we infer that growth in woodfuel production is due to charcoal production specifically, and this is supported by multiple findings in our analysis. The number and percentage of charcoal using households increased, and we observed comparative decreases in purchased firewood use (Figs. 2 and 3), suggesting wider availability of charcoal in marketplaces over the study period. Second, our modeling analysis showed increased odds of using charcoal among households that live near key protected areas, demonstrating that geographic proximity to charcoal production (and related cash generating livelihood opportunities) may increase access (Figs. 4 and 5). Together, these findings support that a robust charcoal sector in Malawi has quickly grown and increased charcoal's accessibility to rural households.

Purchasing power has been cited as a major barrier to household energy transitions in Malawi and similar contexts in SSA (Guta et al., 2022; Puzzolo et al., 2016; Wassie et al., 2021). Charcoal is more expensive than purchased firewood (Alinaitwe & Bergland, 2024; Zulu, 2010), so the increased popularity of charcoal signals that more households are willing and able to allocate financial resources to household fuel. This finding is important to the broader energy transitions landscape, as increased charcoal purchasing among rural populations suggests potential for successful introduction of other energy and fuel commodities, such as liquid petroleum gas, in the future. In Malawi, previous research provides evidence of households' willingness to pay for improved cooking technologies, further supporting the notion that alternative fuels may be successfully introduced to rural populations (Jagger & Jumbe, 2016).

Rural households with few members, headed by males and young individuals, and that are asset rich but land-poor have greater odds of using charcoal as their primary fuel compared to collected firewood (Fig. 4). Consistent with other studies on rural land tenure in Malawi and sub-Saharan Africa (Anseeuw et al., 2016; Jayne et al., 2022; Lindsjö et al., 2021), we observe a 45% decline in median land area planted per household, from 1.8 acres in 2004 to 1.0 acre in 2020 (Table 1). The population density of rural areas steadily grew (Table 2), decreasing the land available for younger generations (Lindsjö et al., 2021). These trends suggest that a growing cohort of newly formed, younger, and smaller households must seek alternatives to farming livelihoods.

Table 3
Forest cover extent and loss in Malawi's protected areas, 2001–2020.

Protected Area	Region	Total area (km^2)	Forest extent 2000 (km^2)	Forest loss (km^2)					
				2001–05	2006–11	2012–17	2018–20	Total	% loss
South Viphya FR*	North	1425.24	1132.10	43.31	78.96	201.76	66.18	390.21	34.47
Dzalanyama FR†	Centre	976.95	381.82	1.24	7.53	33.38	39.62	81.77	21.42
Nyika NP	North	3092.29	1412.84	19.18	8.74	10.90	2.77	41.59	2.94
Mulanje Mountain FR*	South	473.00	236.98	5.90	8.11	11.86	11.19	37.06	15.64
Lengwe NP†	South	928.20	169.81	4.53	15.76	13.30	0.98	34.57	20.36
Liwonde FR†	South	275.66	172.90	0.46	0.64	11.64	18.75	31.49	18.21
Zomba-Malosa FR*	South	189.84	118.78	2.59	6.54	12.01	8.04	29.18	24.56
Namizimu FR	South	797.30	571.46	3.06	6.13	14.10	5.49	28.78	5.04
Mchese FR†	South	88.90	61.17	0.68	0.87	8.07	4.36	13.98	22.83
Chimaliro FR†	North	163.02	83.26	0.02	0.03	0.78	9.57	10.40	12.49
Mangochi FR	South	448.40	350.35	0.22	0.65	4.61	2.39	7.87	2.25
Matandwe FR	South	285.98	74.45	1.23	1.06	3.07	0.85	6.21	8.34
Musisi FR†	North	68.10	43.45	0.92	0.37	1.56	3.24	6.09	14.00
Kaning'ina FR	North	175.73	161.72	1.61	1.58	1.15	1.72	6.06	3.74
Mwabvi GR*	South	344.34	61.76	1.60	1.68	1.29	0.45	5.02	8.12
Perekezi FR	North	176.88	103.78	0.13	0.22	2.03	1.72	4.10	3.95
Kasungu NP	Centre	2358.63	450.61	0.64	1.17	1.68	0.26	3.75	0.83
Nkhotakota GR*	Centre	1784.37	895.61	0.28	0.54	1.85	0.43	3.10	0.35
Mchinji FR	Centre	181.77	82.57	0.25	0.59	1.13	0.75	2.72	3.29
Mua-Livulezi FR	Centre	118.36	56.56	0.13	0.16	0.92	1.07	2.28	4.02
Mtangatanga FR	North	51.76	37.78	0.10	1.14	0.37	0.44	2.05	5.41
Vwaza GR*	North	980.69	347.18	0.37	0.42	0.23	0.32	1.34	0.39
Dedza-Salima Escarpment*	Centre	278.31	105.87	0.10	0.09	0.46	0.68	1.33	1.25
Mafinga Hills FR	North	41.57	8.28	0.20	0.21	0.16	0.05	0.62	7.63
Thuma FR	Centre	161.76	55.15	0.04	0.04	0.18	0.00	0.26	0.46
Liwonde NP	South	505.67	40.69	0.13	0.07	0.04	0.00	0.24	0.59

Table ordered by total forest area lost, 2001–2020.
† Key protected areas where > 10% of the forest cover area in 2000 was lost between 2001 and 2020, and no formal woodfuel extraction activities nor major fires have been documented; mapped in Figs. 6 and 7.
* Forest reserves with timber plantations. FR = Forest reserve; NP=National park; GR = Game reserve.

Consequently, they will shift and diversify livelihoods to earn cash incomes through piecework and spend less time and labor on agriculture. Aspects of agrarian life, including tending to crops and collecting firewood for household cooking, may be replaced by purchasing household necessities, such as food and charcoal for cooking, when markets are accessible. Finally, our finding that households headed by younger individuals have greater odds of charcoal use suggests a shift in fuel preferences along generational lines. Policymakers and private sector actors may consider promoting alternative fuels to younger individuals that are more likely to use charcoal and may be willing to adopt "newer" fuels and technologies.

Relatedly, households who rely on subsistence agriculture may choose to ease the burden of firewood by using charcoal on an infrequent basis as a supplementary fuel. We found that large households had greater odds of supplementary charcoal use compared to no charcoal use, which contrasts with our finding that smaller, land-poor households had greater odds of primary charcoal use (Fig. 5). More research is needed to understand the nuances of fuel stacking dynamics in rural households. Specifically, due to the limitations of our cross-sectional dataset, we could not assess whether supplementary charcoal use leads to primary charcoal use, but this is another pathway through which rural households may transition to primary charcoal use that should be investigated in future studies.

5.3. Implications for natural resource management

From an environmental perspective, a scale-up in charcoal production may have large, negative implications for the environment. Informal charcoal producers typically cut entire trees for carbonization, resulting in greater disturbance to the environment than subsistence firewood collection of branches and deadwood (Zulu & Richardson, 2013). However, the impact of charcoal production on forests and land cover change is difficult to quantify, due to charcoal production co-occurring with other land clearing activities such as agricultural extensification (Branch et al., 2022). Still, hot spots of forest loss at local and regional scales have been directly attributed to charcoal production previously (Bailis et al., 2015; Chidumayo & Gumbo, 2013; Sedano et al., 2020, 2021, 2022). Our analysis of forest loss patterns in protected areas is informed by the authors' observations of fuelwood production occurring in Malawi (particularly in Dzalanyama Forest Reserve) and previous literature which found that forest reserves and protected areas have been major sources of woodfuel production for several decades (Kambewa et al., 2007).

Maps of Dzalanyama, Liwonde, Mchese, and Chimaliro demonstrate striking patterns that are consistent with unsanctioned woodfuel extraction and other woodfuel-intensive activities, where loss encroaches into the centers of the reserves over time (Fig. 7). These patterns show concentrated loss in large swaths of the protected areas. This is contrasted with mosaic patterns of loss throughout much of Lengwe NP and on the Mozambique side of the national border near Dzalanyama FR; these mosaic loss patterns are consistent with land clearing for smallholder agriculture.

Forest loss in Dzalanyama has been previously linked to charcoal production for household energy use (M. F. Gondwe et al., 2019; Katumbi et al., 2017; Munthali & Murayama, 2011, 2015; Zulu, 2010). Forest loss in Liwonde has been linked to woodfuel extraction for tobacco curing and brick making (Chibwana et al., 2013) and is the site of a long-standing community-forestry co-management program (Chilongo, 2014; Jumbe & Angelsen, 2011; Puzzolo et al., 2016). There is no previous documentation of Mchese as a site of major forest loss or degradation in the literature, though it is considered part of Mulanje Mountain Forest, which has been documented as a site of charcoal production (Zulu, 2010). Chimaliro FR, Musisi FR, and Lengwe NP have not been documented as sites of major forest loss previously. Finally, we note that our estimates of forest loss and degradation align with trends observed in other studies; we found that rates of deforestation and forest degradation accelerated substantially post 2009 (Skole et al., 2021). In Dzalanyama, Munthali & Murayama, 2011 observed immense

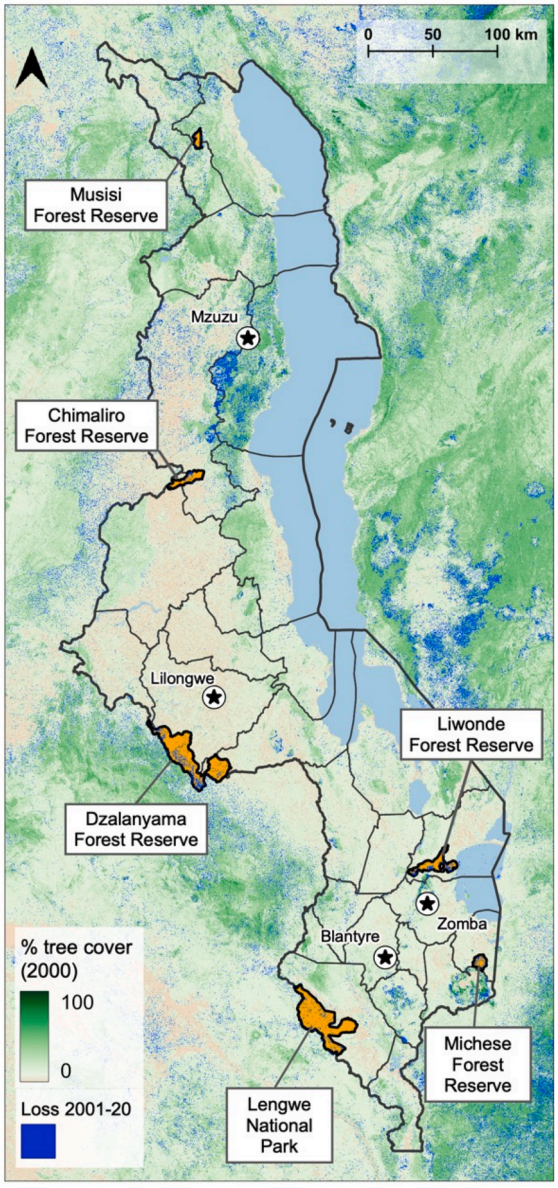


Fig. 6. Locations of key protected areas with high rates of forest loss in Malawi. Key protected areas are federally managed protected areas where > 10% of the forest cover area in 2000 was lost between 2001 and 2020, and no formal woodfuel extraction activities nor major fires have been documented. Malawi's national and district boundaries are represented in black; urban centers are noted on the map. Tree cover loss in Malawi 2001–2020 represented in blue, based on the Global Forest Cover Dataset (Hansen et al., 2013). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

deforestation from 2000 to 2008. Though their estimates of forest extent and deforestation do not align with our estimated total area of loss, the temporal trends in deforestation and forest degradation are similarly extensive.

While our analysis cannot explicitly link expansion of charcoal use in Malawi to forest loss in these specific reserve areas, several factors support the inference that a large portion of observed forest loss in key protected areas is due to increased woodfuel production. First, we do not find any association between forest loss around EAs and fuel use in our modeling analysis (Figs. 4 and 5); we infer that woodfuels must therefore be sourced from other geographies. Malawi's charcoal sector is almost entirely domestic, with limited import or export across the shared borders with Mozambique, Tanzania, and Zambia (Kambewa et al., 2007). Second, in our models, primary charcoal use was strongly associated with household proximity to key protected areas (Figs. 4 and 5), suggesting that rural households near these sites may earn cash from production and transport wages and access the charcoal easily. Third, land clearing for agricultural extensification in protected areas is unlikely and therefore points to unsanctioned woodfuel extraction for charcoal production and other woodfuel-intensive activities, such as tobacco curing and brick making. Fourth, in Dzalanyama reserve, forest loss has been contained within Malawi and has not reached westward to the border of Mozambique, suggesting that most of the forest loss is by domestic actors expanding deeper into the reserve area as woody biomass stocks are depleted over time. This is consistent with what other studies have found based on interviews with relevant stakeholders (Kambewa et al., 2007; Zulu, 2010). In a 2015 study of rural residents near Dzalanyama in particular, 40% noted that the primary driver of forest loss and degradation in the reserve was charcoal production, followed by firewood production (32%), infrastructure development (13%) and illegal timber extraction (11%) (Katumbi et al., 2017). Finally, Tyukavina et al. (2022) estimate that only 0.3% of forest area loss in Malawi between 2000 and 2020 was attributable to forest fires, further suggesting that human activities are the primary driver of loss.

We acknowledge that due to data limitations in our study, forest loss and degradation patterns cannot be explicitly attributed to charcoal production; loss patterns may also be attributable to other woodfuel intensive activities, such as tobacco curing and brick making. Still, our results show that there is an urgent need to delineate the relative contribution of woodfuel extraction for charcoal and design appropriate mitigation strategies that support rural livelihoods and protect natural resources. Though our analysis emphasizes the link between woodfuels for cooking and forest loss, rural population growth is certainly driving woodfuel extraction for brick burning, as bricks are a major building material for rural dwellings (Ngwira & Watanabe, 2019). Furthermore, the acceleration of forest loss in key protected areas is consistent with increases in number and percentage of households using charcoal over time (Fig. 3, Table 3). These broad trends represent a massive scaling up of forest degradation and tree loss that should be closely monitored and investigated to identify the specific drivers of forest loss in each area.

5.4. Study limitations and strengths

There are limitations to this work due to data availability and study design. We are unable to attribute the relative contribution of forest loss to common rural activities, such as curing tobacco or brick making, that are also woodfuel intensive (Ngwira & Watanabe, 2019). As discussed above, remotely sensed forest cover data does not allow us to attribute the cause of woody biomass loss. Additionally, while the IHS survey instrument included detailed questions about livelihood strategies, there is not enough detail to directly link each of woodfuel intensive activity with land cover change, nor with individual households. While we did not find reports of forest fires or other activities in these areas, it is possible that some forest loss is attributable to fires or sanctioned land clearing for other purposes. Relatedly, while we carefully selected a 10-kilometer radius for our proximity-based variables (forest cover, forest cover loss, and proximity to protected areas), the design of the IHS geospatial offset may have introduced measurement error that obscured truly present associations between features of EA localities and household fuel choices. Future studies may consider refining proximity-based associations to identify association sensitivities at varying distances.

We are limited in our analysis of forest loss and cannot analyze forest regrowth because we use the GFCD. The GFCD presents several analytical advantages (publicly available, global coverage, fine spatial resolution, temporally explicit information about loss), but data on forest gain were not available on an annual basis. Therefore, we could not incorporate values of time-varying biomass stocks at the local level

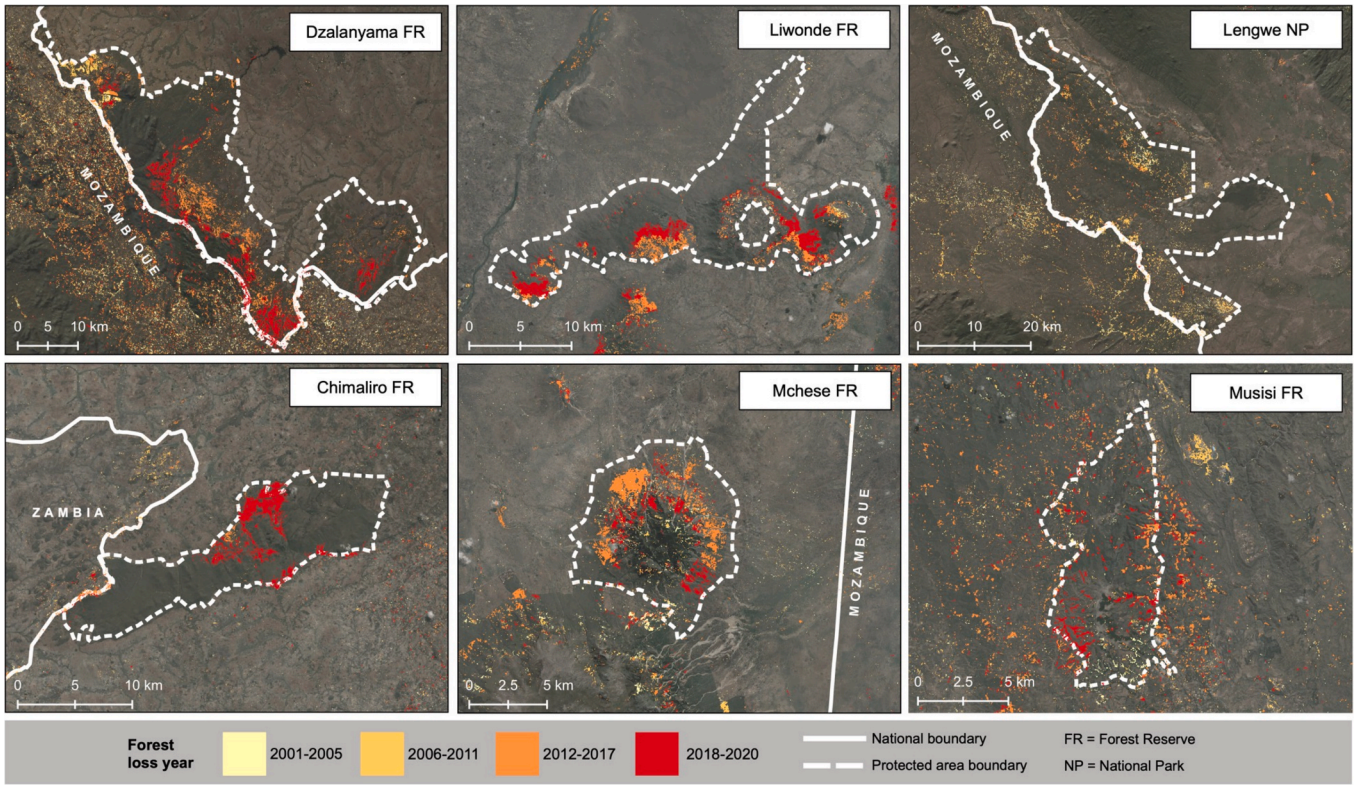


Fig. 7. Patterns of tree cover loss in key protected areas, 2001–2020. Key protected areas are federally managed protected areas where > 10% of the forest cover area in 2000 was lost between 2001 and 2020, and no formal woodfuel extraction activities nor major fires have been documented. Tree cover loss data are from the Global Forest Cover Dataset (Hansen et al., 2013).

nor account for forest regrowth in protected areas during our study period. However, between 2000 and 2020, Malawi gained 42 kilohectares of forest cover nationally (Potapov et al., 2022). This level of gain over a 20-year period is minimal, representing approximately 3% of year 2000 baseline forested area.

Finally, we analyze a time series of cross-sectional survey data, which does not allow us to observe changes and transitions within individual households. Despite this, the ability to observe national trends over a 16-year interval allows us to generate great insights into changes in several interrelated demographic, economic, and environmental factors. We leverage geospatial information included in the IHS to integrate high-quality measures of environmental change over varying geographies in the study area, allowing us to generate new insights and a nuanced understanding of the ongoing energy transition in Malawi.

6. Conclusions

Charcoal use as a primary cooking fuel is underexplored in the energy transitions literature. Until very recently, research and policy analysis has typically lumped charcoal into the same category as firewood (Branch et al., 2022; Mwampamba et al., 2013; Rose et al., 2022). In SSA, where charcoal production and use are increasing in both rural and urban settings, this is a serious omission in our understanding of energy transitions in Africa, particularly rural energy transitions. The energy transitions literature is heavily dominated by a focus on high-and middle-income country settings, or in the case of low-income country settings, focused on transitions from traditional woodfuels to modern fuels. Our analysis contributes to both the energy transitions literature and the poverty-environment dynamics literature. By leveraging nationally representative survey and geospatial datasets, we estimate associations between demographic, economic, and land cover factors and rural cooking fuel consumption from 2004 to 2020 in Malawi, which we argue is a highly representative case study that is comparable to other rural populations in sub-Saharan Africa.

A significant finding of this study is that by 2020 an estimated 25.0% of Malawian households were using charcoal as a primary or supplementary fuel, demonstrating that charcoal should be given more policy and research attention. Furthermore, the rate of charcoal use among rural households increased substantially, from 3.9% to 13.4%, demonstrating that charcoal's importance is growing among the rural population, which for some time will be the majority population in SSA. High levels of forest cover loss in forest reserves that are documented woodfuel production sites suggest that substantial forest loss is linked to the growth of rural populations and a developed charcoal sector. Approximately 20% of forest cover in Dzalanyama, Liwonde, and Mchese Forest Reserves have been lost, highlighting localities where interventions to protect forests and support rural livelihoods are urgently needed.

Our findings imply two important areas of continued research and policy action to ensure a sustainable energy transition in Malawi and similar contexts in sub-Saharan Africa. First, observed growth in purchased woodfuels suggests an opportunity for the promotion of alternative fuels in urban areas and rural areas affected by population pressure, where we find evidence of increased woodfuel purchasing. If households are willing and able to allocate expenditures towards energy needs, the success of other fuel markets, such as liquid petroleum gas, is more possible (Osiolo, 2017; Perros et al., 2024; Rahut et al., 2016). This shift is especially significant in rural areas, where households have traditionally relied on collected woodfuel for household energy needs. Now may be the time to seriously consider promotion of alternative fuels in rural areas, especially to households we identified as most likely to use charcoal who are smaller, younger, asset-rich, and land-poor.

Second, with respect to livelihoods, consequences of rural population pressure will continue to shift land resources available to households. As a result, alternative livelihoods to agriculture will be highly sought after. The woodfuel sector will continue to develop in Malawi with a sustained

demand among a growing urban population and, as our analysis shows, a changing rural population. The importance of the charcoal sector to rural livelihoods is already acknowledged by the Government of Malawi's (2017) integrated National Charcoal Strategy (NCS) 2017–2027. While the NCS integrates economic development and sustainable management of natural resources, the dynamics of land access in rural areas with high population growth should be incorporated into the design of alternative livelihoods interventions.

CRediT authorship contribution statement

Katerina E.B. Beach: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Varun Goel:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Yu Wu:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Barbara Entwisle:** Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis, Conceptualization. **Pamela Jagger:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data (Appendix A) to this article can be found online at https://doi.org/10.1016/j.worlddev.2026.107542. Appendix A includes Table A1, Table A2, Table A3, Table A4, and Figure A1.

Data availability

Our public replication package ican be found on GitHub.

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